

Antonio Franco

linkedin.com/in/Antonio-J-Franco | Full-time co-op availability (Jan 2027–Dec 2028) | Open to relocation

Mechatronics student who enjoys building across hardware and software. Drawn to unconventional products engineered for efficiency, longevity, maintainability, and the people who use them.

Education

Bachelor of Science in Engineering – Minnesota State University, Mankato

August 2026 - December 2028

Certificate of Achievement in Engineering – Ventura College

May 2026

Projects

Portfolio: francoaj.com

3-DOF Robotic Leg

- Designed and assembled 3-DOF, quadruped-style prototype robotic leg with mixed actuation mechanisms including gear-driven joint, planetary gearbox, and linear actuator linkage system
- Implemented and solved inverse kinematics system in MicroPython to compute joint angles and generate end-effector motion along step-shaped curves using a control loop
- Built Blender models with forward and inverse kinematics armatures to compare against simulated behavior
- Modeled linear actuator and linkage geometry by measuring extension vs. rotation cycles and approximating joint angle mapping as a linear function for simpler control input conversion
- Integrated off-the-shelf motor control and power electronics to focus development on mechanical design, kinematics, and motion planning

Axial Flux Alternator

- Performed research to decide on specifications of the alternator
- Designed mechanical components in Onshape for 3D printing and revised parts as they were tested
- Designed and built 4-speed mechanical tool to keep count of windings and speed up manual assembly

Automotive Maintenance

- Perform routine vehicle maintenance including oil changes, tire rotations, and spark plug replacement
- Replaced mechanical components, including suspension drop links and valve cover gaskets

Air Quality Monitor

- Integrated an off the shelf sensor, microcontroller, and epaper display
- Modeled, sliced, 3D printed, and assembled a sturdy sensor housing without fasteners
- Set up Wi-Fi access point on a server with a database table and live dashboard that the sensor logs data to
- Developed data processing script to calculate and record 10-minute mean of sensor readings

QMK Mechanical Keyboard

- Researched and selected components to design the keyboard from scratch
- Designed the case in CAD to 3D print and made the wiring diagram of the key matrix
- Wrote and compiled custom QMK firmware

Raspberry Pi IoT & Networked Systems Platform

- Maintaining self hosted services with a Tailscale Tailnet and local network Wi-Fi access point
- Docker, Tailscale, SQLite (with CRUD server)
- Networking, self-hosting, automation

Skills

Programming & CAD: Python, SQLite, Supabase, Onshape, Blender

Embedded: Microcontrollers, Linux Administration

Fabrication: Soldering, 3D Printing